Prime Geometry III:
A Generalized Bertrand Principle for Prime Constellations

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Abstract

In PG II we introduced the Twin-Prime Bertrand Postulate (TPB): every interval $(x, 2x]$ with $x \geq 11$ contains a twin prime pair. Here we show that the same dyadic inequality appears to hold, with the same universal factor 2, for every admissible pair-constellation. We formulate the Generalized Bertrand Principle (GBP):

$$\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1 \qquad (x \geq P^*_{\mathcal{C}})$$

for every admissible constellation $\mathcal{C} = \{0, g\}$ with Hardy–Littlewood singular series $\mathfrak{S}(\mathcal{C}) > 0$. We give a generalized $2P$-beats lemma linking (GBP) to the Prime Triangle angle geometry, prove the equivalence between the dyadic and ratio formulations, and extract a data-driven envelope

$$G^{\mathcal{C}}_j \ < \ 0.171\,(\log P^{\mathcal{C}}_j)^{3.22}$$

that fits all three constellations with $R^2 \geq 0.98$. Empirical verification uses the twin ($g = 2$), cousin ($g = 4$) and sexy-prime ($g = 6$) sequences up to 10^{10}: 27 412 679 twins, 27 409 999 cousins and 54 818 296 sexy pairs, with $P^*_{\{0,2\}} = 11$, $P^*_{\{0,4\}} = 7$, and $P^*_{\{0,6\}} = 5$. We discuss the asymptotic interpretation, the drift of the twin-gap exponent β toward the Hardy–Littlewood value 2, and the apparent universality of the overshoot law $\Omega(T) \sim (\log\log T)^{\sim 1}$.

1 Introduction

PG II isolated a single inequality — the Twin-Prime Bertrand Postulate — as a natural target in twin-prime distribution:

$$\pi_2(2x) - \pi_2(x) \geq 1 \qquad (x \geq 11).$$

It was verified there for $x \leq 10^9$; we have since extended the verification to $x \leq 10^{10}$, still without exception beyond the trivial small-x case $T_k = 5$. The inequality has an equivalent geometric reading: every angle-record of the Prime Triangle sequence with $p_n \geq 3$ is realized by a twin prime.

The factor 2 in (TPB) is a Bertrand factor — it matches the dyadic interval $(x, 2x]$ — but until now we had not asked whether it was specific to twins. In this note we investigate the behavior of other prime constellations (the cousin primes, $\{0, 4\}$, and the sexy primes, $\{0, 6\}$) and find that the same bound holds:

For every admissible pair-constellation, the dyadic inequality $\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1$ holds for all sufficiently large x, with a threshold $P^*_{\mathcal{C}}$ that is very small in every case examined.

We call this the Generalized Bertrand Principle (GBP) and establish its equivalence with an elementary ratio form and, via a generalized $2P$-beats lemma, with a family of angle-record statements in the Prime Triangle sense. We verify the principle for all three constellations to 10^{10}, and combine the data into a uniform empirical envelope for the extreme gaps.

Throughout, $\mathbb{P}$ denotes the primes. The Hardy–Littlewood prime-tuple conjecture predicts, for any admissible constellation $\mathcal{C}$, an asymptotic twin-like density $\pi_{\mathcal{C}}(x) \sim 2\,\mathfrak{S}(\mathcal{C})\,x/(\log x)^2$, with $\mathfrak{S}(\mathcal{C})$ the singular series. For the three constellations studied here, $\mathfrak{S}(\{0,2\}) = \mathfrak{S}(\{0,4\}) = C_2 \approx 0.6602$ and $\mathfrak{S}(\{0,6\}) = 2C_2$ (see Section **??**).

2 Admissible Constellations and (GBP)

Definition 1 (Admissible pair-constellation)**.** *A* pair-constellation *is a set $\mathcal{C} = \{0, g\}$ with $g \geq 2$ even. It is* admissible *if for every prime q the reduction $\mathcal{C} \pmod q$ does not cover all residue classes. Equivalently, the singular series*

$$\mathfrak{S}(\mathcal{C}) = \prod_{q \text{ prime}} \frac{1 - \nu_{\mathcal{C}}(q)/q}{(1 - 1/q)^{|\mathcal{C}|}}$$

is positive, where $\nu_{\mathcal{C}}(q)$ is the number of residues modulo q occupied by $\mathcal{C}$.

For every admissible $\mathcal{C} = \{0, g\}$, let

$$P_1^{\mathcal{C}} < P_2^{\mathcal{C}} < \cdots$$

be the sequence of leading members: $P_j^{\mathcal{C}} \in \mathbb{P}$ with $P_j^{\mathcal{C}} + g \in \mathbb{P}$.

Definition 2 ($\mathcal{C}$-counting function)**.** $\pi_{\mathcal{C}}(x) = \#\{p \leq x : p \in \mathbb{P},\, p + g \in \mathbb{P}\}$.

Conjecture 3 (Generalized Bertrand Principle, (GBP))**.** *For every admissible pair-constellation $\mathcal{C} = \{0, g\}$ there exists a threshold $P_{\mathcal{C}}^*$ such that*

$$\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1 \qquad (x \geq P_{\mathcal{C}}^*).$$

Equivalently, $P_{j+1}^{\mathcal{C}} < 2\,P_j^{\mathcal{C}}$ for every j with $P_j^{\mathcal{C}} \geq P_{\mathcal{C}}^*$.

Proposition 4 (Dyadic and ratio forms)**.** *For each admissible $\mathcal{C}$ with fixed threshold $P_{\mathcal{C}}^*$, the two formulations above are equivalent.*

Proof. The proof is identical to Proposition 2.1 of PG II. If $P_{j+1}^{\mathcal{C}} < 2P_j^{\mathcal{C}}$ whenever $P_j^{\mathcal{C}} \geq P_{\mathcal{C}}^*$, fix $x \geq P_{\mathcal{C}}^*$ and let $P_j^{\mathcal{C}}$ be the largest term $\leq x$; then $P_{j+1}^{\mathcal{C}} < 2P_j^{\mathcal{C}} \leq 2x$, giving at least one $\mathcal{C}$-pair in $(x, 2x]$. Conversely, if the dyadic inequality holds and $P_j^{\mathcal{C}} \geq P_{\mathcal{C}}^*$, the dyadic form with $x = P_j^{\mathcal{C}}$ produces $P_{j+1}^{\mathcal{C}} \in (P_j^{\mathcal{C}}, 2P_j^{\mathcal{C}}]$; strict inequality follows because $2P_j^{\mathcal{C}}$ is even and $P_{j+1}^{\mathcal{C}}$ is odd (for $g > 0$ even, both members of the pair are odd). $\square$

Observe that (TPB) is the special case $\mathcal{C} = \{0, 2\}$ with $P_{\{0,2\}}^* = 11$.

3 A Generalized $2P$-Beats Lemma

Recall from PG I the Prime Triangle angle of a pair (p, q) with $p < q$:

$$\alpha(p, q) = \arctan(p/q), \qquad \rho(p, q) = p/q.$$

For a fixed constellation $\mathcal{C} = \{0, g\}$ and a member $P_j^{\mathcal{C}}$, the pair $(P_j^{\mathcal{C}}, P_j^{\mathcal{C}} + g)$ has angle $\alpha_j^{\mathcal{C}} = \arctan(P_j^{\mathcal{C}}/(P_j^{\mathcal{C}} + g))$.

Lemma 5 (Generalized $2P$-beats)**.** *Let $\mathcal{C} = \{0, g\}$ and $\mathcal{D} = \{0, h\}$ with $g \neq h$, and let $(P, P + g)$, $(Q, Q + h)$ be prime pairs for $\mathcal{C}$ and $\mathcal{D}$ respectively. Then*

$$\rho(P, P + g) > \rho(Q, Q + h) \iff h\,P > g\,Q.$$

In particular:

(i) For $h = 2$, $g = 4$: a cousin pair $(P, P + 4)$ beats a twin pair $(Q, Q + 2)$ in angle iff $P > 2Q$ (the $2P$-beats rule of PG II).

(ii) For $h = g$: both pairs belong to the same constellation $\mathcal{C}$, and $\rho(P, P + g) > \rho(Q, Q + g) \iff P > Q$. The angle sequence within a fixed $\mathcal{C}$ is monotonic in the leading prime.

(iii) For a constellation $\mathcal{C} = \{0, g\}$ and its "doubled companion" $\mathcal{C}' = \{0, 2g\}$, a $\mathcal{C}'$-pair at P beats a $\mathcal{C}$-pair at Q iff $P > 2Q$. The factor 2 is universal for the $\mathcal{C} \leftrightarrow \mathcal{C}'$ comparison, independent of g.

Proof. $\frac{P}{P+g} > \frac{Q}{Q+h} \iff P(Q + h) > Q(P + g) \iff hP > gQ$. The specializations follow. □

Part (iii) gives the geometric explanation for the universal factor 2 in (GBP): for every admissible $\mathcal{C} = \{0, g\}$, the *doubled-gap* companion $\mathcal{C}' = \{0, 2g\}$ is also admissible (its singular series is positive), and the two constellations are related by the same $2P$-beats threshold. The dyadic factor of 2 in the Bertrand inequality is the natural density counterpart of the same factor in the angle comparison.

4 The Generalized Angle-Record Theorem

For twins, PG II's Angle-Record Theorem states that every record-setting α_n among consecutive prime pairs (p_n, p_{n+1}) with $p_n \geq 3$ is a twin pair, and that this is equivalent to (TPB).

To extend this to a general $\mathcal{C}$ we need two things. First, a *domain* on which $\mathcal{C}$-pairs can plausibly dominate the angle-records: if $\mathcal{C}$ has gap $g > 2$, twins always beat $\mathcal{C}$-pairs of comparable leading prime (since $\rho(P, P + g) < \rho(P, P + 2)$), and twin records preempt $\mathcal{C}$-records in the unrestricted sequence.

Fix $\mathcal{C} = \{0, g\}$ and consider only the pairs $(p, p + g')$ with $g' \geq g$ and both entries prime — call these *$\mathcal{C}$-admissible pairs*. Within this restricted universe, the $2P$-beats threshold for a competing pair of strictly larger gap $g' > g$ is $P > (g'/g)Q \geq (g+2)/g \cdot Q$, so the *binding* competing threshold is the companion $\mathcal{C}' = \{0, g + 2\}$ with factor $(g + 2)/g$. Only in the special case $g = 2$ (twins, with companion cousins) is this binding factor equal to 2.

Definition 6 ($\mathcal{C}$-angle-record)**.** *A $\mathcal{C}$-admissible pair $(P^*, P^* + g^*)$ with $g^* \geq g$ is a $\mathcal{C}$-angle-record if $\rho(P^*, P^* + g^*) > \rho(P, P + g')$ for every earlier $\mathcal{C}$-admissible pair $(P, P + g')$.*

Theorem 7 (Generalized Angle-Record Theorem)**.** *For each admissible $\mathcal{C} = \{0, g\}$, the following are equivalent:*

(i) $P^{\mathcal{C}}_{j+1} < \frac{g+2}{g} P^{\mathcal{C}}_j$ *for every j with $P^{\mathcal{C}}_j \geq P^{\star}_{\mathcal{C}}$ (for some effective $P^{\star}_{\mathcal{C}}$ depending on the gap).*

(ii) Every $\mathcal{C}$-angle-record with $P^* \geq P^{\star}_{\mathcal{C}}$ is a pair of constellation $\mathcal{C}$.

For $g = 2$, $(g + 2)/g = 2$ and (i) reduces to (TPB)*; Theorem* **??** *is then PG II's Angle-Record Theorem. For $g = 4$ the factor is* 1.5*; for $g = 6$ the factor is* 4/3*; and so on.*

Proof sketch. The argument is analogous to the twin case. (i)⇒(ii): suppose (i) holds. Between two consecutive $\mathcal{C}$-members $P^{\mathcal{C}}_j$ and $P^{\mathcal{C}}_{j+1}$, a competing $\mathcal{C}$-admissible pair $(P, P + g')$ with $g' > g$ beats the $\mathcal{C}$-record at $P^{\mathcal{C}}_j$ iff $P > (g'/g)P^{\mathcal{C}}_j$; the binding case is $g' = g + 2$, requiring $P > ((g + 2)/g)P^{\mathcal{C}}_j$. Under (i), $P^{\mathcal{C}}_{j+1} < ((g + 2)/g)P^{\mathcal{C}}_j$, so no competing pair in the window can exceed the threshold, and the next $\mathcal{C}$-admissible record is either a larger-P pair of gap g (i.e., $P^{\mathcal{C}}_{j+1}$) or none until then; meanwhile $P^{\mathcal{C}}_{j+1}$ does set a new record because $\rho(P^{\mathcal{C}}_{j+1}, P^{\mathcal{C}}_{j+1} + g) > \rho(P^{\mathcal{C}}_j, P^{\mathcal{C}}_j + g)$.

(ii)⇒(i): contrapositive. If (i) fails for some j, there exist $(P, P + g')$ with $g' > g$, both primes, and $P^{\mathcal{C}}_j < P \leq P^{\mathcal{C}}_{j+1}$ with $P > ((g+2)/g)P^{\mathcal{C}}_j$. By Lemma **??**, this pair beats the previous $\mathcal{C}$-record, so a non-$\mathcal{C}$-record appears. The three smallest anomalies (very small P) are handled by direct inspection, producing the effective $P^{\star}_{\mathcal{C}}$. □

5 Empirical Verification up to 10^{10}

All data in this section derive from a single segmented (odd-only) sieve of the primes up to 10^{10}, followed by a streaming extraction of all pairs $(p, p + g)$ with both primes for $g \in \{2, 4, 6\}$. Cross-boundary pairs are handled by a three-entry buffer. The sieve confirms 455 052 511 primes below 10^{10} and the following constellation counts:

constellation $\mathcal{C}$	gap g	count up to 10^{10}
twin	2	27 412 679
cousin	4	27 409 999
sexy	6	54 818 296

5.1 Ratio envelopes

For each constellation we tabulate $\sup r^{\mathcal{C}}_k$ on tails $P^{\mathcal{C}}_j > X$:

scope	twin sup r	cousin sup r	sexy sup r
all	2.200000	2.333333	1.571429
$P_j > 100$	1.280374	1.283465	1.224299
$P_j > 10^3$	1.081880	1.109790	1.051647
$P_j > 10^6$	1.000711	1.000865	1.000695
$P_j > 10^9$	1.000004	1.000004	1.000002

The three envelopes converge to within 10^{-5} of 1 by $P_j > 10^9$, with comparable decay rates.

5.2 Bertrand threshold $P^*_{\mathcal{C}}$

The top ratios in each constellation (ranks 1–5 shown):

constellation	P_j	P_{j+1}	r_j	gap
twin	5	11	2.2000	6
twin	17	29	1.7059	12
twin	3	5	1.6667	2
twin	11	17	1.5455	6
twin	41	59	1.4390	18
cousin	3	7	2.3333	4
cousin	19	37	1.9474	18
cousin	7	13	1.8571	6
cousin	43	67	1.5581	24
cousin	13	19	1.4615	6
sexy	7	11	1.5714	4
sexy	5	7	1.4000	2
sexy	17	23	1.3529	6
sexy	23	31	1.3478	8
sexy	13	17	1.3077	4

Violations of the strong $r < 2$ bound:

constellation	# violations of $r_j < 2$	$P^*_{\mathcal{C}}$
twin	1 (only $(5,7) \to (11,13)$)	11
cousin	1 (only $(3,7) \to (7,11)$)	7
sexy	0	5

This verifies (GBP) with the stated thresholds for all three constellations up to 10^{10}. In particular, GBP holds with the *same* factor 2 across constellations; the user's suggestion of larger factors (3 for cousins, 4 for sexy) is true but unnecessarily loose — the $r < 2$ bound captures the structure exactly.

5.3 Per-decade twin overshoot

The twin overshoot $\Omega(T) = \max G_k/(\log T^\star_k)^2$ per decade (stable regime $10^3 \le T_k < 10^9$):

decade	max G_k	at T_k	$(\log T)^2$	Ω
10^3	210	5 879	75.33	2.788
10^4	630	62 297	121.87	5.169
10^5	1 452	850 349	186.42	7.789
10^6	1 722	9 923 987	259.55	6.635
10^7	2 868	96 894 041	338.16	8.481
10^8	4 770	698 542 487	414.71	11.502
10^9	6 030	4 289 385 521	491.93	12.258

6 A Refined Envelope Conjecture

For each constellation we compute $\sup(r_j - 1)$ over tails $P_j^{\mathcal{C}} > T_{\min}$ at $T_{\min \in \{10^2, 10^3, 10^5, 10^7, 10^9\}}$:

$T_{\min}$	twin $\sup(r-1)$	cousin $\sup(r-1)$	sexy $\sup(r-1)$
10^2	2.80×10^{-1}	2.83×10^{-1}	2.24×10^{-1}
10^3	8.19×10^{-2}	1.10×10^{-1}	5.16×10^{-2}
10^5	5.10×10^{-3}	4.33×10^{-3}	3.34×10^{-3}
10^7	1.43×10^{-4}	1.49×10^{-4}	9.87×10^{-5}
10^9	3.86×10^{-6}	4.02×10^{-6}	2.26×10^{-6}

The three envelopes decay at essentially the same rate. A weighted least-squares fit of the log-linear form

$$\log\big((r_j - 1)T_j\big) = \log C_{\mathcal{C}} + \delta_{\mathcal{C}} \log\log T_j$$

on the five $T_{\min}$ points yields:

constellation	$\delta_{\mathcal{C}}$	$C_{\mathcal{C}}$	R^2
twin	3.2884	0.1645	0.9970
cousin	3.2209	0.2013	0.9957
sexy	3.1538	0.1507	0.9931
pooled (uniform fit)	3.2210	0.1709	0.9824

The fitted exponents agree across constellations within 4.2% (range $[3.15, 3.29]$), strongly suggesting a common asymptotic δ. The constants $C_{\mathcal{C}}$ vary by roughly 29% (range $[0.15, 0.20]$); this larger spread is driven by the cousin envelope, whose leading offenders at small P_j (dominated by the pair $(3,7) \to (7,11)$) inflate its fitted prefactor relative to the asymptotic regime.

The pooled fit — obtained by concatenating all fifteen data points and fitting a single (C, δ) pair — yields

$$r_j - 1 \;\le\; 0.171\,\frac{(\log P_j)^{3.22}}{P_j}, \qquad G_j^{\mathcal{C}} \;<\; 0.171\,(\log P_j^{\mathcal{C}})^{3.22} \qquad (R^2 = 0.982). \tag{1}$$

Conjecture 8 (Refined envelope, data-driven)**.** *For every admissible pair-constellation $\mathcal{C} = \{0, g\}$ there exist constants $C_{\mathcal{C}}, \delta_{\mathcal{C}}$ with $\delta_{\mathcal{C}} \to \delta^*$ as g varies over admissible gaps, for some universal $\delta^* \in [3.15, 3.30]$, such that*

$$G_j^{\mathcal{C}} \;=\; P_{j+1}^{\mathcal{C}} - P_j^{\mathcal{C}} \;<\; C_{\mathcal{C}}\,(\log P_j^{\mathcal{C}})^{\delta_{\mathcal{C}}} \qquad (P_j^{\mathcal{C}} \ge P_{\mathcal{C}}^*).$$

*The pooled uniform envelope (**??**) fits all three examined constellations with $R^2 \ge 0.98$.*

This is strictly stronger than (GBP), which is recovered by the coarse bound $G_j^{\mathcal{C}}/P_j^{\mathcal{C}} \to 0$. It is compatible with the Hardy–Littlewood prediction for the *typical* gap $G_j^{\mathcal{C}} \sim (\log P)^2/(2\mathfrak{S}(\mathcal{C}))$: the exponent $\delta \approx 3.2 > 2$ captures the *extreme* gap, which exceeds the typical gap by an overshoot factor that itself grows slowly with P.

7 Asymptotic Interpretation

7.1 Exponent drift toward $\beta = 2$

In PG II the twin-gap power law $G_k \approx A(\log T_k)^\beta$ was fitted on primes up to 10^9, yielding $\beta \approx 1.866$ stable across sub-ranges. Extension to 10^{10} shifts the fits upward:

fit range $T_k >$	β (10^9 data)	β (10^{10} data)
10^3	1.8633	1.8865
10^5	1.8659	1.8872
10^7	1.8664	1.8889
10^9	—	1.8957

The drift is monotone; each fit band grew $\Delta\beta \approx +0.02$ across the new decade. Under the interpretation that the true asymptotic exponent is $\beta = 2$ (Hardy–Littlewood), our observations are consistent with a slow finite-size convergence whose pace is controlled by the overshoot factor Ω.

7.2 Overshoot growth law

Fits of the per-decade twin overshoot $\Omega(T)$ on the seven stable points yield:

model	form	R^2
(a)	$\Omega = A(\log T)^b$, $A = 0.147$, $b = 1.43$	0.9133
(b)	$\log \Omega = a + b(\log\log T)^\alpha$, $\alpha = 0.69$	0.9135
(c)	$\Omega = D(\log T)^\gamma$, $\gamma = 0.10$	0.8964
(d)	$\Omega = A + B\log\log T$, $A = -17.48, B = 9.32$	0.8959

Models (a) and (b) are essentially tied, indicating $\Omega \sim (\log T)^{1.43}$ or equivalently $\log\Omega \sim (\log\log T)^{0.7}$. This is a *Cramér–Granville*-type growth: Cramér conjectured $p_{n+1} - p_n = O((\log p)^2)$ for ordinary primes, refined by Granville to include a $\log\log p$ factor. For twin-prime gaps we see an analogous correction with an exponent $\alpha \approx 0.7$ in the $\log\log$ factor, slightly sub-logarithmic but firmly positive.

7.3 Why factor 2 is natural and universal

The factor 2 in (GBP) has a clean interpretation: it is the dyadic Bertrand factor. For ordinary primes, Bertrand's postulate ensures a prime in every $(x, 2x]$; for any admissible $\mathcal{C}$, the expected count of $\mathcal{C}$-pairs in $(x, 2x]$ under Hardy–Littlewood is

$$\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \sim 2\mathfrak{S}(\mathcal{C})\,\frac{x}{(\log x)^2},$$

which diverges to infinity. Unconditionally we do not know that this count is ≥ 1 for every sufficiently large x, but the data indicate that the inequality becomes binding extremely early ($P^*_{\mathcal{C}}$ at most the first few pair members in every case we examined) and is maintained thereafter with orders of magnitude of slack.

Lemma **??**(iii) provides the geometric explanation: the factor 2 is exactly the $2P$-beats threshold for the doubled-companion relation $\mathcal{C} \leftrightarrow \mathcal{C}' = \{0, 2g\}$. In the angle picture, moving from a $\mathcal{C}$-member at Q to a $\mathcal{C}'$-member at P requires $P > 2Q$ to beat the $\mathcal{C}$-angle; this threshold is universal in g. The dyadic factor of 2 in Bertrand's inequality is the density counterpart.

7.4 Why violations occur only at tiny P

The near-misses of the bound $r < 2$ all lie at $P_j \leq 881$ across all three constellations; past this, no ratio in our data exceeds 1.1. The reason is purely statistical: at small P, the singular series effectively constrains the positions of constellation pairs, and a small neighborhood of primes may fail to contain both p and $p+g$. Once P is large enough that the expected count in $(P, 2P]$ exceeds 1 by many standard deviations, fluctuations cannot drive it to zero. The Hardy–Littlewood estimate gives the expected count $\sim 2\mathfrak{S}(\mathcal{C})P/(\log P)^2$, which equals 10 by $P \sim 10^4$ (twin/cousin) or $P \sim 10^3$ (sexy); a factor of 10 is already too large for statistical fluctuations to explain a complete absence.

8 Conclusion

We have identified the Generalized Bertrand Principle — a conjectural dyadic inequality $\pi_{\mathcal{C}}(2x) - \pi_{\mathcal{C}}(x) \geq 1$ with a universal factor of 2 — as the natural extension of PG II's twin-prime Bertrand postulate to every admissible pair-constellation. The principle is equivalent to the simple ratio condition $P_{j+1}^{\mathcal{C}} < 2P_j^{\mathcal{C}}$ and is supported geometrically by a generalized $2P$-beats lemma expressed in the Prime Triangle angle. We have verified the principle in three constellations (twins, cousins, sexy primes) up to $P = 10^{10}$, giving $P^*_{\{0,2\}} = 11$, $P^*_{\{0,4\}} = 7$, $P^*_{\{0,6\}} = 5$.

Three observations flow from the computation:

1. The cousin-prime count at 10^{10} (27 409 999) is within 0.01% of the twin-prime count (27 412 679), confirming the Hardy–Littlewood prediction of identical densities for $\{0, 2\}$ and $\{0, 4\}$.

2. The ratio envelopes tighten at essentially the same rate across constellations. Fitted exponents $\delta_{\mathcal{C}}$ agree within 4.2% across the three constellations (range $[3.15, 3.29]$); a pooled uniform fit gives $G_j^{\mathcal{C}} < 0.171(\log P)^{3.22}$ with $R^2 = 0.982$.

3. The twin-gap exponent β drifts upward from 1.866 (at 10^9) to 1.896 (at 10^{10} restricted to $T_k > 10^9$), consistent with the asymptotic value $\beta = 2$ predicted by Hardy–Littlewood; concurrently the overshoot factor $\Omega(T)$ grows roughly like $(\log T)^{1.4}$ or $(\log\log T)^{0.7}$.

Future work. A conditional proof of (GBP) under Hardy–Littlewood is immediate along the same lines as Theorem 5.1 of PG II. An unconditional proof would require a short-interval result $\pi_{\mathcal{C}}(x + y) - \pi_{\mathcal{C}}(x) > 0$ at dyadic scale $y = x$, which is strictly beyond current sieve-theoretic capabilities (bounded-gaps results of Zhang, Maynard, and Polymath8 give such short-interval existence only at some scales, not all). We expect that the Maynard multi-dimensional sieve, adapted to dyadic windows, is the natural apparatus for an attack on GBP.

The apparent universality of the envelope across constellations, and the unified overshoot law, suggest that the Generalized Bertrand Principle is part of a larger picture: HL-admissible constellations behave as coordinated copies of the twin-prime sequence, scaled by their singular-series constants, and subject to a common growth law for their extreme gaps. Quantifying this universality conjecturally — in particular, obtaining a closed form for the asymptotic overshoot rate — is the natural next step.

Data availability

All computations reproduce from `scripts/pg_structural_1e10.py` and `scripts/pg_extend_1e10.py`. Saved datasets (`twins_1e10.npy`, `cousins_1e10.npy`, `sexy_1e10.npy`, and associated CSVs) are stored under the `data/` directory.

Acknowledgements

This paper is the third in a series. PG I introduced the Prime Triangle construction and derived geometric quantities; PG II established the Twin-Prime Bertrand Postulate and the Angle-Record Theorem; the present note generalizes those results to all admissible pair-constellations.